

Question 01 [2+2 Marks]

a) Represent the following **polynomial** using **Polynomial library**. Print the value of the Polynomial function at $x=8$.

$$f(x) = (x - 9)^2 + 7x + x^3$$

b) Find out the value of $\frac{f(x) + f''(x)}{(f'''(x))^2}$ at $x = 3$? Round up the value to 5 decimal place

Question 02 [4+2 Marks] You are an intern at a race car company, and you are analyzing the speed profile of a prototype car during its initial test runs. The engineers recorded the following data points:

Time x	Speed f(x)
0	0
3	30
4	40
7	45

SET - A

Your supervisor needs you to predict the car's speed at 6 seconds to verify performance against design goals.

However, instead of using the regular Lagrange interpolation, you are instructed to use a new form of Lagrange polynomial interpolation:

General form of the **new n degree lagrange polynomial**:

$$p_n(x) = \sum_{k=0}^n f(x_k) l_k(x)$$

where

$$l_k(x) = \prod_{j=0, j \neq k}^n \frac{x_k - x_j}{x_k - x}$$

a) Evaluate the Lagrange bases $[l_0(x), l_1(x), l_2(x), l_3(x)]$ of the polynomial that goes through the above data points given in the table. Take $x = 2$

b) Calculate the P(x) from Lagrange Interpolating polynomial at $x = 2$.

#Driver Code

x = [0, 3, 4, 7]

y = [0, 30, 40, 45]

eval_x = 2

Lagrange_Basis = calc_lagrange_basis(x, eval_x)

print(Lagrange_Basis)

print(f"\nInterpolated value P({eval_x}) = {P_x}")

Question 03 [2 + 3 = 5 Marks]

Suppose there is a function,

$$f(x) = 2x^3 - 5x^2 + 3x + 7$$

To analyze the behavior of the Absolute Error when approximating the derivative of the function at $x = 2$ using **Backward Difference**, you have to **determine the minimum value of h** that ensures the absolute error stays below a given **tolerance threshold** of 0.0025. Additionally, **calculate how many reductions of h** are required to meet this condition.

As you already know,

$$1. \text{ Backward Difference, } f'(x) = \frac{f(x) - f(x-h)}{h}$$

$$2. \text{ Absolute error} = |\text{Actual derivative} - \text{Approximate derivative}|$$

You should start with an **initial step size of h = 0.4** and iteratively **reduce h by half**, if necessary, until the error condition is satisfied.

Hint: You have to use **Polynomial class** from the **numpy** library to define the function, $f(x)$ and Your output should clearly state the **minimum h** and the **total number of reductions** required to achieve the desired accuracy.

#Driver Code

error(f, f_prime, h, x, tolerance)

Question.01 [2+2 Marks]

a) Represent the following polynomial using **Polynomial library**. Print the value of the Polynomial function at $x=8$.

$$f(x) = (x-9)^2 + 7x - x^2 + x^3 = x^3 - 18x + 81 + 7x - x^2 + x^3 = 2x^3 - x^2 - 11x + 81$$

b) Find out the value of $\frac{f(x) + f''(x)}{(f'(x))^2}$ at $x = 3$? Round up the value to 5 decimal place

Question 02 [4+2 Marks] You are an intern at a race car company, and you are analyzing the speed profile of a prototype car during its initial test runs. The engineers recorded the following data points:

Time x	Speed f(x)
0	0
3	20
5	90
7	85

Your supervisor needs you to predict the car's speed at 6 seconds to verify performance against design goals.

However, instead of using the regular Lagrange interpolation, you are instructed to use a new form of Lagrange polynomial interpolation:

General form of the new n degree lagrange polynomial:

$$p_n(x) = \sum_{k=0}^n f(x_k) l_k(x)$$

where

$$l_k(x) = \prod_{j=0, j \neq k}^n \frac{2 * x_k - x_j}{3 * x_k - x}$$

SET - B

a) Evaluate the Lagrange bases $[l_0(x), l_1(x), l_2(x), l_3(x)]$ of the polynomial that goes through the above data points given in the table.

Take $x = 2$.

b) Calculate the $P(x)$ from Lagrange Interpolating polynomial at $x = 2$.

#Driver Code

$x = [0, 3, 5, 7]$

$y = [0, 20, 90, 85]$

$eval_x = 2$

$Lagrange_Basis = calc_lagrange_basis(x, eval_x)$

$print(Lagrange_Basis)$

$print(f"\n Interpolated value P(\{eval_x\}) = \{P_x\}")$

Question 03 [2 + 3 = 5 Marks]

Suppose there is a function,

$$f(x) = 2x^3 - 5x^2 + 3x + 7$$

To analyze the behavior of the Absolute Error when approximating the derivative of the function at $x = 2$ using **Forward Difference**, you have to **determine the minimum value of h** that ensures the absolute error stays below a given **tolerance threshold of 0.0025**. Additionally, **calculate how many reductions of h** are required to meet this condition.

As you already know,

$$1. \text{ Forward Difference, } f'(x) = \frac{f(x+h) - f(x)}{h}$$

$$2. \text{ Absolute error} = |\text{Actual derivative} - \text{Approximate derivative}|$$

You should start with an **initial step size of $h = 0.4$** and iteratively **reduce h by half**, if necessary, until the error condition is satisfied.

Hint: You have to use **Polynomial class** from the **numpy** library to define the function, $f(x)$ and Your output should clearly state the **minimum h** and the **total number of reductions** required to achieve the desired accuracy.

#Driver Code

$error(f, f_prime, h, x, tolerance)$